

Digital Material Passport

ID DMP-METAL-005 - Version 1.0.0  
Issue Date 2025-05-17 - Certificate Type EN 10204 3.1

| Manufacturer                                                                                         | Customer                                                                                              | Business Transaction |                                           |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------|
| ACME Metal Works GmbH<br>Industrial Park 123<br>52066 Aachen<br>DE<br>quality@acme-metal.example.com | European Research Institute<br>Science Boulevard 42<br>75015 Paris<br>FR<br>materials@eri.example.org | Order                | PO-23456 · 1000 kg<br>Pos. 1 · 2025-05-01 |
|                                                                                                      |                                                                                                       | Delivery             | DN-65432 · 1000 kg<br>Pos. 1 · 2025-05-16 |

Product

|                   |                              |              |                                     |
|-------------------|------------------------------|--------------|-------------------------------------|
| Product Name      | Titanium Alloy Ti-6Al-4V ELI | Number (UNS) | R56401                              |
| Batch ID          | T-65432-01                   | Shape        | RoundBar - D 30 mm - Length 3000 mm |
| Surface Condition | Machined                     | Product Norm | ASTM F136 (2022)                    |
| Weight            | 1000 kg                      |              |                                     |
| Production Date   | 2025-05-15                   |              |                                     |
| Country of Origin | DE                           |              |                                     |

Chemical Analysis

Heat Number T-65432 - Melting Process VAR - Casting Method VacuumCasting - Casting Date 2025-05-14 - Sample Location Product

| Symbol | Ti    | Al             | V              | Fe   | O    | C     | N     | H   |
|--------|-------|----------------|----------------|------|------|-------|-------|-----|
| Unit   | %     | %              | %              | %    | %    | %     | %     | ppm |
| Min    |       | 5.50           | 3.50           |      |      |       |       |     |
| Max    |       | 6.50           | 4.50           | 0.25 | 0.13 | 0.080 | 0.050 | 120 |
| Actual | 89.32 | 6.02<br>± 0.05 | 3.95<br>± 0.03 | 0.18 | 0.11 | 0.026 | 0.012 | 35  |

Mechanical Properties

| Property                  | Actual      | Minimum | Maximum | Method                  | Status  |          |
|---------------------------|-------------|---------|---------|-------------------------|---------|----------|
| <b>Fatigue Test</b>       |             |         |         | ASTM E466               | ✓       |          |
| Room temperature, R = 0.1 |             |         |         |                         |         |          |
| Cycles (N)                | 10000       | 50000   | 100000  | 500000                  | 1000000 | 10000000 |
| Value                     | 650         | 635     | 610     | 590                     | 570     | 550      |
| <b>Notch Sensitivity</b>  | 0.85 - 0.92 |         |         | Internal Method TS-5432 | ✓       |          |

|                   |             |                         |   |
|-------------------|-------------|-------------------------|---|
| Notch Sensitivity | 0.85 - 0.92 | Internal Method TS-5432 | ✓ |
|-------------------|-------------|-------------------------|---|

Physical Properties

| Property | Actual            | Target/Min | Maximum | Method    | Status |
|----------|-------------------|------------|---------|-----------|--------|
| Density  | 4.43 ± 0.01 g/cm³ | 4.43 g/cm³ |         | ASTM B311 | ✓      |

Supplementary Tests

| Test                       | Result / limits                                      | Method                          | Status |
|----------------------------|------------------------------------------------------|---------------------------------|--------|
| Microstructure Examination | Equiaxed alpha with intergranular beta               | ASTM E407                       | ✓      |
| Ultrasonic Inspection      | Yes - No indications greater than reference standard | ASTM E2375                      | ✓      |
| Surface Quality Assessment | Class 1 - Medical Grade                              | Visual Inspection per ASTM F136 | ✓      |
| Alpha Case Depth           | 5 µm - max 25 µm                                     | Microhardness Traverse          | ✓      |
| Grain Size Distribution    | 8 - 10 ASTM No. - target 7 - 12 ASTM No.             | ASTM E112                       | ✓      |

Hardness Profile (ASTM E384), values in HV

|                            |     |     |     |     |     |     |     |     |
|----------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
| Distance from surface (mm) | 0.1 | 0.5 | 1   | 2   | 3   | 5   | 10  | 15  |
| Value [HV]                 | 345 | 350 | 352 | 350 | 349 | 348 | 351 | 347 |

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM F136 and meets all requirements for surgical implant applications.

Validated by **Dr. Markus Weber**, Head of Metallurgy, Research & Quality - 2025-05-17